

Adrián Hernández Padrón

Senior Data Scientist | AI Engineer

📍 Tenerife, Spain ✉️ adrianhdezp10@gmail.com ☎️ +34 611 45 14 44 🔗 adrihp.dev 🌐 [adrianhernandezpadron](https://adrianhernandezpadron.com)

🔗 [Adrihp06](#)

Profile

Senior Data Scientist and AI Engineer designing and delivering complete AI products. Independently design and develop Travel Ruter, from the frontend and backend to its conversational agent, retrieval workflows and MCP integrations. Lead the Data Science team at Intelequia, with professional experience in applied ML, optimization and MLOps on Azure, Databricks and Microsoft Fabric.

Travel Ruter

Travel Ruter | Product design and full-stack AI development

Sept 2025 - present

My own travel-planning product, designed and built end to end: interactive maps, daily itineraries, budgets, saved research and an AI assistant operating on live trip data.

- Own product design and implementation across the React/TypeScript frontend, FastAPI backend, asynchronous SQLAlchemy persistence and PostgreSQL/PostGIS database.
- Designed a PydanticAI conversational agent with multi-provider LLM support, streaming responses and multi-turn, state-aware workflows to research destinations, plan itineraries and edit persisted trip records.
- Built retrieval-augmented generation (RAG) over saved trip notes and research, combining scoped evidence with live web search and source citations to ground planning and writing tasks.
- Created internal and remote MCP servers, exposing typed trip-management, discovery, routing and scheduling tools. Integrated Travel Ruter with ChatGPT through authenticated MCP access, with scoped permissions and server-side validation of actions.
- Developed context and capability selection using JEV, explicit record identity, freshness checks and additive tool loading to reduce repeated model work and preserve prompt-cache reuse.
- Designed and ran controlled comparative evaluations of agent architectures, measuring response quality, correctness of saved changes, latency and inference cost. [Technical study](#).
- Built and maintain the containerized application with Docker Compose, automated checks and agent tracing; use evaluation results and failure analysis to guide changes to the agent architecture.

Professional Experience

Intelequia, Senior Data Scientist

Jan 2026 - present

- Lead the Data Science team across modeling, Azure and Databricks architecture, and MLOps practices for model versioning, monitoring and deployment.
- Build GenAI agents and data agents on Azure AI Foundry, defining tool schemas, prompts, grounding on client data and evaluation sets to validate changes before release.
- Designed and implemented an end-to-end ML solution to assign materials, quantities and prices to construction work items based on installation descriptions, covering database connectivity, data processing, model training and registration, a custom UI for interacting with the ML system, containerization and application deployment.

DISA Corporación Petrolífera S.A., Data Scientist

Sept 2023 - Jan 2026

- Developed cluster-constrained route optimization models for logistics planning.

- Delivered GenAI use cases end to end: sentiment analysis, a RAG assistant over internal documents and LLM workflows for manual review tasks, including retrieval configuration, prompt iteration and answer validation.
- Built repeatable training and inference workflows with Docker, Kubernetes and ArgoCD; introduced MLflow on Databricks for experiment tracking and model lifecycle management.
- Led a joint project with the University of La Laguna on anomaly detection in authentication logs, comparing embeddings and sequential models.

Loro Parque Foundation - Universidad de la Laguna, Research Data Scientist Jan 2023 - Aug 2023

- Developed a Sound Event Detection (SED) system combining inverse Fourier transform signal processing with machine learning to detect acoustic events in marine recordings.
- Applied YAMNet, a deep neural network based on MobileNetV1, to classify detected sound events in recordings of different species.

Turismo de Tenerife, Junior Data Scientist July 2022 - Jan 2023

- Performed predictive analysis of tourist arrivals to Tenerife by nationality.
- Maintained the tourism Data Warehouse and supported statistical analysis of tourism data.

Education

MSc Universidad Internacional de Valencia (VIU), Astronomy and Astrophysics June 2025

MSc Universidad Internacional de Valencia (VIU), Data Science and Big Data June 2023

BSc Universidad de La Laguna (ULL), Physics Sept 2021

Technical Skills

AI and agents: PydanticAI, MCP servers and integrations, RAG, web research, tool calling, context management, prompt caching; OpenAI, Anthropic, Vertex AI, Azure AI Foundry, LangGraph, LangChain

Machine learning and MLOps: Python, SQL, PyTorch, scikit-learn, YAMNet, embeddings, optimization, MLflow, Databricks, Microsoft Fabric, Apache Spark, Docker, Kubernetes, ArgoCD, CI/CD

Full-stack engineering and evaluation: FastAPI, Pydantic, async SQLAlchemy, PostgreSQL/PostGIS, React, TypeScript, Mapbox, pytest; multi-turn agent evaluation, persisted-state checks, Logfire, latency and inference-cost analysis

Certifications

- Designing and Implementing Multi-Agent AI Solutions (AI-500), Microsoft (2026) - Exam passed
- Microsoft Certified: Machine Learning Operations Engineer Associate (AI-300)
- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)

Languages

Spanish: native. English: professional working proficiency.